Constrained damping for looped two-stream reasoners

Working note

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Abstract

We study exactly invertible two-stream iterations, the *kick* (conformal) layer and the *relax* layer, as weight-tied looped reasoners.

- Without constraints on the damping, the models we trained overthink. The standard block and relax drift: the state grows linearly in the iteration count, and the re-injected input is washed out. Kick with small fixed rates (0.005) is bounded in principle, but behaves the same way for about 1000 iterations.
- In relax the drift has an exact mechanism. A common coordinate integrates the forces, so it drifts linearly exactly when the running average of the forces has a nonzero limit, which we observe in the trained models.
- Two constraints make the linear part a strict contraction while preserving exact inversion and the constant volume rate: a *common-mode damping* $\delta > 0$ for relax, and *positivity* of the kick rates (bounded away from zero). Both produce attractors: the shift-free spectral radius is below 1, and after 1024 iterations 79–100% of trajectories sit at a fixed point; the rest are on short cycles.
- An attractor need not encode the answer, however. With supervision beyond the training horizon (a detached carry), both constrained couplings keep their ListOps accuracy to 128 iterations (relax with $\delta = 1/4$: $0.944 \to 0.946$, three seeds), where every unconstrained model collapses. On 3-hop only the constrained kick keeps it ($0.962 \to 0.946$); damped relax with carry loses 9% of its correct answers.

The experiments are small ($d = 32$, one layer per iteration, 1–3 seeds).

1 Two-stream iterations

The state is $s = (L, H) \in \mathbb{R}^{T\times d} \times \mathbb{R}^{T\times d}$, for T tokens. Let $e \in \mathbb{R}^{T\times d}$ be the token and position embedding, re-injected at every iteration. The branches are the pre-LN sublayers

$$A(H) = \mathrm{Attn}\big(\mathrm{LN}(H + e)\big), \qquad M(L) = \mathrm{MLP}\big(\mathrm{LN}(L + e)\big). \tag{1}$$

We write $a = e^{-\gamma}$, $b = e^{-\alpha}$ (diagonal, the retained fractions) and $c = e^{-\delta} \in (0, 1]$. One iteration $\Phi : s \mapsto s^+$ is one of

$$\textbf{kick:} \quad L^+ = a \odot \big(L + A(H)\big), \qquad H^+ = b \odot H + M(L^+), \tag{2}$$

$$\textbf{relax}_\delta\textbf{:} \quad L^+ = a \odot L + (1 - a) \odot \big(cH + A(H)\big), \qquad H^+ = b \odot H + (1 - b) \odot \big(cL^+ + M(L^+)\big). \tag{3}$$

Plain relax is $\delta = 0$, i.e. $c = 1$. The kick map is the conformal reversible layer of [1], iterated with tied weights. The prediction after r iterations is $\mathrm{softmax}\big(W\,\mathrm{LN}(H_r)\big)$, with $s_0 = (e, e)$.

Proposition 1 (Exact inverse, constant volume)**.** *Both maps are bijections, with explicit inverses*

$$H = b^{-1} \odot \big(H^+ - (1 - b) \odot (cL^+ + M(L^+))\big), \qquad L = a^{-1} \odot \big(L^+ - (1 - a) \odot (cH + A(H))\big)$$

for relax$_\delta$; the kick inverse is analogous. For every state,

$$\log |\det D\Phi(s)| = -T \sum_{i=1}^{d} (\gamma_i + \alpha_i),$$

independently of δ.

Proof. L^+ is affine in L, with a coefficient independent of the state, and H^+ is affine in H given L^+. So $D\Phi$ is a product of two block-triangular matrices with diagonal blocks $\mathrm{diag}(a)$ and $\mathrm{diag}(b)$. The parameter c only enters the off-diagonal blocks. □

Training. Each step draws $r \sim \mathcal{U}\{r_{\text{lo}}, \dots, r_{\text{hi}}\}$ and takes the loss $\ell\big(y, \Phi^r(s_0)\big)$ at the last iteration only. With the *detached carry* [2], with probability 1/2 the graded iterations instead start from

$$\tilde{s} = \text{sg}\left(\Phi^{r_0}(s_0)\right), \qquad r_0 \sim \mathcal{U}\{1, \dots, 64\}, \tag{4}$$

where sg stops the gradient. This is the loss $\ell\big(y, \Phi^r(\tilde{s})\big)$. It supervises states up to $r_0 + r_{\text{hi}} = 72$ iterations deep while differentiating through at most $r_{\text{hi}} = 8$, at the cost of $\mathbb{E}[r_0]/2 \approx 16$ forward-only iterations per step on average.

2 Why unconstrained loops do not settle

Lemma 2 (The common coordinate integrates the forces)**.** *For relax with $\delta = 0$, define*

$$C = \frac{a(1-b) \odot L + (1-a) \odot H}{1-ab}, \qquad \kappa = \frac{(1-a)(1-b)}{1-ab}.$$

Then, exactly,

$$C^+ = C + \kappa \odot \big(A(H) + M(L^+)\big),$$

while the difference $L - H$ is contracted by ab when the forces vanish.

Proof. Substitute (3) with $c = 1$ into $(1-ab)C^+ = a(1-b)L^+ + (1-a)H^+$ and collect terms. □

So $C_r = C_0 + \kappa \odot \sum_{k<r} v_k$ with $v_k = A(H_k) + M(L_{k+1})$: nothing restores C, and fixed points require the forces to cancel exactly. Telescoping gives a conditional drift statement.

Corollary 3 (Linear drift)**.** *If the running average $\bar{v}_r = \frac{1}{r}\sum_{k<r} v_k$ converges to some $v \neq 0$, then $C_r/r \to \kappa \odot v$.*

The architecture permits this but does not force it. At large norm the pre-LN branches are 0-homogeneous in the state, since $\text{LN}(\lambda H + e) \to \text{LN}(H)$ as $\lambda \to \infty$, so the forces depend only on the normalised direction of the state. They may still rotate or cancel on average. In the trained relax models the premise holds (Fig. 3): $\|P\bar{v}_r\|$ approaches a nonzero limit (4.5 on ListOps and 1.9 on 3-hop at $r = 2048$), and $\cos(v_r, \bar{v}_r) \to 1.00$. The looped standard block $x \mapsto x + A(x) + M(x + A(x))$ telescopes in the same way, with $\kappa = 1$.

Drift has a second consequence. The input enters only through $\text{LN}(H + e)$, so its relative weight

$$\omega_r = \frac{1}{T}\sum_t \frac{\|Pe_t\|}{\|P(H_{r,t} + e_t)\|} \tag{5}$$

falls like $1/r$ when $\|PH_r\|$ grows linearly, and the iteration forgets its input. Here P removes each stream's per-token channel mean.

Remark 4 (LayerNorm shift symmetry and the quotient). Every branch and the readout are invariant to per-token shifts $s\mathbf{1}$ of their stream argument. With *channelwise* retention this does not make P a valid quotient: in kick, shifting L by $s\mathbf{1}$ changes L^+ by $s\,a$, whose centred part $s\,Pa$ is nonzero unless a is constant across channels (for relax_δ a shift $(s\mathbf{1}, t\mathbf{1})$ gives $(s - ct)Pa$). The projected Jacobian $\Pi D\Phi\,\Pi$, $\Pi = \text{diag}(P, P)$, then drops a feedback path, and its spectral radius need not bound the full one. All runs reported here use one *scalar* rate per stream. Then $Pa = Pb = 0$, the shift directions form an invariant subspace, $\Pi D\Phi(I - \Pi) = 0$, and $\text{spec}\, D\Phi$ is the union of the quotient spectrum and the per-token linear part K acting on the stream means. For relax with $\delta = 0$, $(s\mathbf{1}, s\mathbf{1})$ is then an exact neutral direction. As a check, we also measure the unprojected radius: it agrees with ρ_r within 0.01 for every model. In the trained models the drift lies almost entirely *outside* the shift directions (Fig. 2).

3 Constraining the damping

Proposition 5 (Linear part)**.** *With the forces switched off ($A = M = 0$), each channel of relax_δ evolves by*

$$K = \begin{pmatrix} a & (1-a)c \\ (1-b)ac & b + (1-a)(1-b)c^2 \end{pmatrix}, \qquad \det K = ab, \quad \text{tr}\, K = a + b + (1-a)(1-b)c^2.$$

For $a, b \in (0, 1)$:

- *if* $c = 1$*, then* $\operatorname{spec} K = \{1, ab\}$*, and the common mode* $(1, 1)$ *is neutral;*
- *if* $c < 1$*, both eigenvalues are real and lie in* $(0, 1)$*, so* $\rho(K) < 1$*.*

For kick, $K = \operatorname{diag}(a, b)$ *and* $\rho(K) = \max(a, b) < 1$ *if and only if* $\gamma, \alpha > 0$*.*

Proof. The characteristic polynomial is $p(\lambda) = \lambda^2 - t\lambda + ab$ with $t = \operatorname{tr} K$. Since $t \geq a + b \geq 2\sqrt{ab}$, the roots are real and positive. Moreover $p(1) = (1 - a)(1 - b)(1 - c^2)$, and the vertex satisfies $t/2 < (1 + ab)/2 < 1$. Hence both roots are smaller than 1 if and only if $c < 1$. □

Corollary 6 (Bounded state, no washout)**.** *Suppose* $\rho(K) < 1$*. The pre-LN branches are bounded, since LayerNorm outputs have bounded norm and attention is a convex combination. Hence* $s^+ = Ks + f(s)$ *with* $\sup \|f\| < \infty$*, and*

$$\sup_r \|s_r\| \leq C_K \Big(\|s_0\| + \frac{\sup \|f\|}{1 - \tilde{\rho}} \Big)$$

for any $\tilde{\rho} \in (\rho(K), 1)$*, where* C_K *accounts for the non-normality of* K*. In particular* ω_r *stays bounded away from zero.*

The constraints change the equilibria. Relax_δ rests at $L_* = cH_* + A(H_*)$ and $H_* = cL_* + M(L_*)$. At $\delta = 0$ this reduces to the rate-independent equilibrium $A(H_*) + M(L_*) = 0$ of the standard block. Kick rests where the forcing balances the leakage: $(e^\gamma - 1)L_* = A(H_*)$ and $(1 - e^{-\alpha})H_* = M(L_*)$.

We use $\delta \in \{1/64, 1/16, 1/4\}$ with $\gamma = \alpha = 3$ (Fig. 4a). For kick we use positive learned rates $\gamma, \alpha = 0.05 + \text{softplus}(\theta)$, initialised at 1. The previously reported kick failures used $\gamma = \alpha = 0.005$, or unconstrained learned rates, which turned negative and diverged. Rates of 0.005 are positive, so the corollary does bound the orbit, but with retention 0.995: an e-folding time of 200 iterations and a state bound of about $a/(1 - a) \approx 200$ times $\sup \|A\|$. Over the horizons we test this looks like drift; the trained model's state saturates only after about 1000 iterations (Fig. 3). What matters is the damping strength relative to the computation horizon, not positivity alone.

Proposition 5 guarantees a bounded orbit, not that the attractor encodes the answer. The loss only sees $r \leq r_{\text{hi}}$, which motivates the carry (4).

4 Experiments

Setup.

- **ListOps** [3]**:** MIN/MAX, nesting depth at most 3, 40 tokens, answer at the last position; training $r \in [2, 8]$, 6000 steps.
- **3-hop pointer chasing:** 8 pair tokens $a \to f(a)$ and 8 queries with targets $f^3(q)$; training $r \in [3, 8]$, 8000 steps.
- **Model:** $d = 32$ per stream, 4 heads, one coupling layer per iteration, about 18k parameters. AdamW with learning rate 10^{-3} and cosine decay, batch size 64.
- **Metrics:** test accuracy at r up to 128; the overthinking rate $O = \Pr[\text{wrong at } 128 \mid \text{correct at } 8]$; the share of changed predictions $\Delta\hat{y} = \Pr[\hat{y}_{128} \neq \hat{y}_8]$, which also counts changes between two wrong labels; the step size $\|\Delta s\|_r = \|s_r - s_{r-1}\|$; and the shift-free spectral radius ρ_r of $P\, D\Phi(s_r)\, P$. We estimate ρ_r by 40 power iterations with Jacobian–vector products and report the median over 64 test sequences.

Findings.

1. **Unconstrained loops drift and overthink.** The standard block and relax grow linearly out to $r = 2048$. Their ω_r decays like $1/r$, and ρ_r exceeds 1 at $r = 8$ and is neutral at $r = 128$ (Figs. 2, 3, 4b). Kick with $\gamma = \alpha = 0.005$ grows for about 1000 iterations, then saturates ($\|Ps\| \approx 1900$ on ListOps) on mostly wrong answers (accuracy 0.20). On ListOps all three lose 72–85% of their correct answers by $r = 128$. On 3-hop the standard block and relax mostly survive (relax: $O = 5.8 \pm 7.0\%$, one seed at 16%), presumably because the answer is formed early and the drift direction preserves it.

	ListOps					3-hop pointer chasing				
model	n	acc@8	acc@128	O [%]	$\Delta\hat{y}$ [%]	n	acc@8	acc@128	O [%]	$\Delta\hat{y}$ [%]
standard block	3	0.918 $\pm$0.015	0.143 $\pm$0.048	85.4 $\pm$5.3	85.1 $\pm$5.0	1	1.000	0.990	1.0	1.0
relax	3	0.917 $\pm$0.018	0.262 $\pm$0.112	72.4 $\pm$12.5	73.3 $\pm$11.1	3	0.990 $\pm$0.014	0.935 $\pm$0.079	5.8 $\pm$7.0	5.8 $\pm$7.0
kick, $\gamma = \alpha = 0.005$	1	0.843	0.193	78.5	80.6	1	0.999	0.155	84.5	84.5
kick, positive rates	1	0.893	0.892	0.7	2.1	1	0.913	0.567	38.8	37.0
relax, $\delta = 1/4$	1	0.866	0.567	36.0	42.2	1	0.991	0.947	4.5	4.5
kick, positive rates + carry	3	0.918 $\pm$0.019	0.916 $\pm$0.021	0.5 $\pm$0.3	1.3 $\pm$0.4	3	0.962 $\pm$0.052	0.946 $\pm$0.071	2.4 $\pm$3.0	2.7 $\pm$3.5
relax, $\delta = 1/4$ + carry	3	0.944 $\pm$0.005	0.946 $\pm$0.005	0.4 $\pm$0.0	1.4 $\pm$0.4	3	0.998 $\pm$0.001	0.909 $\pm$0.024	9.0 $\pm$2.3	9.0 $\pm$2.3

Table 1: Main results, mean $\pm$ s.d. over n seeds (1238–1240), 2000 test sequences per seed. O: share of answers correct at $r = 8$ that are wrong at $r = 128$; $\Delta\hat{y}$: share of predictions that differ between $r = 8$ and $r = 128$. Rows from top: unconstrained baselines; constrained damping; constrained damping with carry.

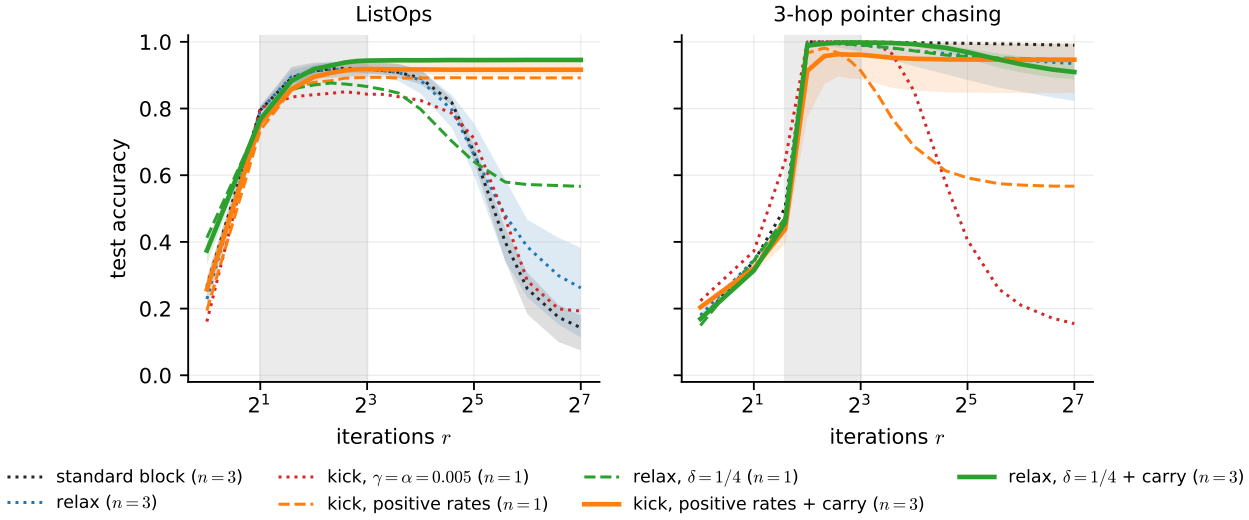


Figure 1: Test accuracy against the number of iterations; grey band: training range. Dotted lines: unconstrained; dashed: constrained damping; solid: constrained damping with carry. Bands show the min–max range over seeds.

2. **Constrained damping produces attractors.** Every constrained variant ends with $\rho_{128} < 1$ (Table 2). At $r = 1024$, 79–100% of test sequences sit at a fixed point (step below 10^{-3}, median about 10^{-6}). The rest are on short cycles, for kick mostly of period 2 (up to 20% of 3-hop sequences with kick and carry). These sequences dominate the mean step $\|\Delta s\|_{128}$ in Table 2.

    - Kick with positive rates learns retentions of 0.25–0.45 per iteration. Among the configurations without carry, it is the only one that keeps its ListOps answers with a bounded state (kick with free learned rates also keeps them, but its state diverges and it collapses on 3-hop). So the coupling is not structurally unstable; the earlier failures were a consequence of vanishing or negative damping.
    - An attractor need not encode the answer. Relax with $\delta = 1/16$ is perfect on 3-hop, yet on ListOps it settles on states whose answers are all wrong at $r = 128$.

3. **Constraint and carry together give accuracy at the horizon and stopping on ListOps; only the constrained kick does so on both tasks** (Table 1, Fig. 5).

    - On ListOps, relax with $\delta = 1/4$ and carry goes 0.944 $\to$ 0.946 ($O = 0.4\%$) and kick with positive rates and carry 0.918 $\to$ 0.916 ($O = 0.5\%$), over three seeds. Only 1.3–1.4% of predictions change between $r = 8$ and 128.
    - On 3-hop, kick with carry holds (0.962 $\to$ 0.946, $O = 2.4 \pm 3.0\%$, one seed at 6.6%). Relax with $\delta = 1/4$ and carry is accurate at $r = 8$ (0.998) but loses $9.0 \pm 2.3\%$ of its correct answers by $r = 128$,

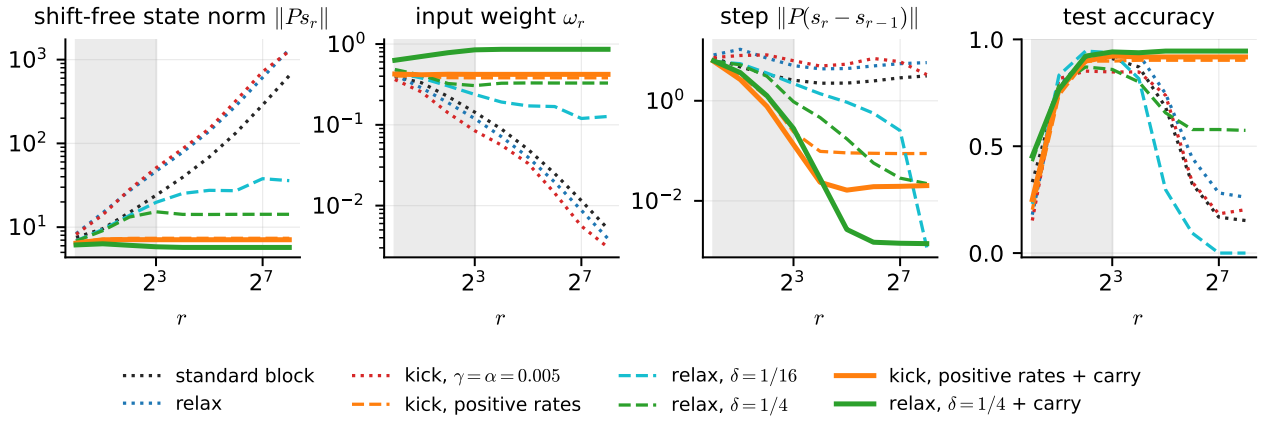


Figure 2: Frozen-checkpoint dynamics on ListOps (seed 1238, 256 test sequences). Left to right: shift-free state norm, input weight ω_r, shift-free step, and accuracy, out to $r = 256$. Unconstrained models drift linearly ($\|Ps_r\| \propto r$) and wash out the input ($\omega_r \propto 1/r$). Constrained models stay bounded, and most of their trajectories converge.

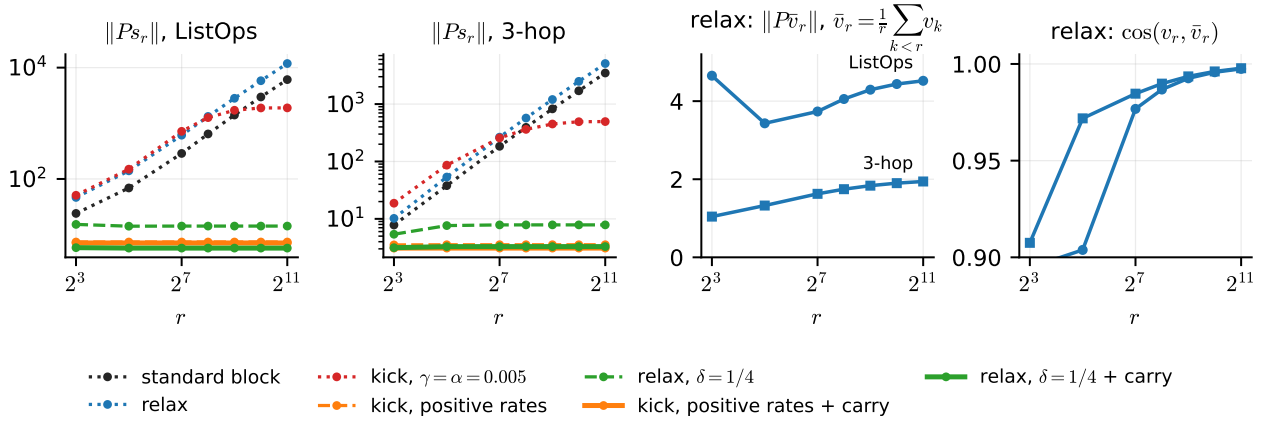


Figure 3: Long horizon (seed 1238, 256 test sequences, $r \leq 2048$). (a, b) Shift-free state norm. The standard block and relax grow linearly; kick with $\gamma = \alpha = 0.005$ saturates after about 1000 iterations (a long transient of a bounded map); constrained models are flat. (c, d) Relax: the running-average force $\bar{v}_r$ tends to a nonzero limit and the current force aligns with it, the premise of Corollary 3.

while settling (step 0.004, $\rho_{128} = 0.94$): again an attractor that is not tied to the answer.

- The carry alone, without a strong contraction, *hurts*: the carried states are washed out, which costs 12–33 points at $r = 8$ (one seed).

4. **A washout fix alone is not enough.** Injecting the input after the LayerNorm keeps ω constant but not the dynamics bounded. It roughly halves overthinking and costs accuracy at the horizon.

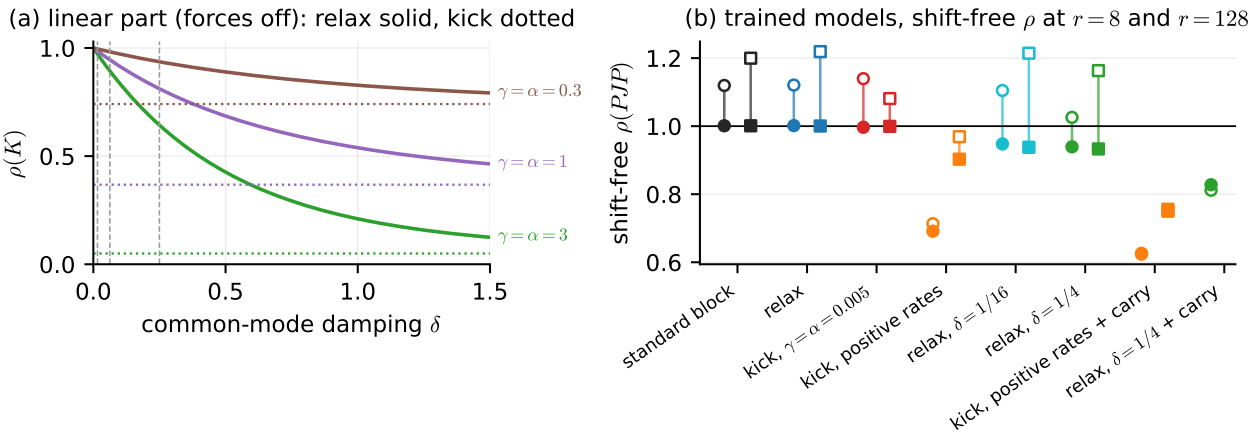


Figure 4: (a) Linear-part spectral radius $\rho(K)$ of relax$_\delta$ (Prop. 5, solid) and kick (dotted), as a function of the common-mode damping; vertical lines mark the values used. (b) Shift-free spectral radius of the trained iteration at $r = 8$ (open markers) and $r = 128$ (filled); circles: ListOps, squares: 3-hop. Unconstrained models are expanding at the training horizon and neutral ($\rho = 1$) far beyond it. Constrained models contract.

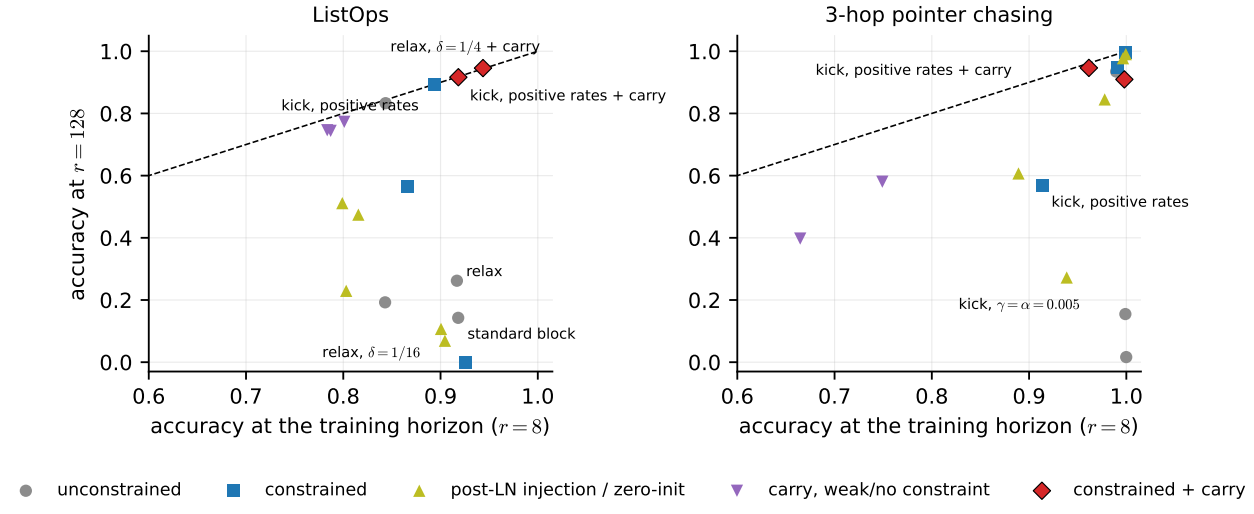


Figure 5: Accuracy at the training horizon against accuracy at $r = 128$ for every configuration in Table 2 (seed means). On the diagonal the two accuracies are equal. Answers can still change, since $\mathrm{acc}_{128} - \mathrm{acc}_8 = G - O'$ with G and O' the shares of all answers that turn from wrong to correct and from correct to wrong; Table 1 reports the share of changed predictions.

	ListOps				3-hop pointer chasing			
model (seed 1238)	acc@8	acc@128	$\|\Delta s\|_{128}$	ρ_{128}	acc@8	acc@128	$\|\Delta s\|_{128}$	ρ_{128}
standard block	0.910	0.179	2.9	1.00	1.000	0.990	1.6	1.00
relax	0.938	0.290	5.6	1.00	0.999	0.988	2.4	1.00
kick, $\gamma = \alpha = 0.005$	0.843	0.193	6	1.00	0.999	0.155	1.2	1.00
kick, free learned rates	0.844	0.833	7e+10	1.20	1.000	0.016	–	–
kick, positive rates	0.893	0.892	0.11	0.69	0.913	0.567	0.022	0.89
relax, $\delta = 1/64$	0.904	0.069	1.2	0.98	0.999	0.996	0.24	0.98
relax, $\delta = 1/16$	0.926	0.000	0.26	0.94	0.999	0.997	0.014	0.93
relax, $\delta = 1/4$	0.866	0.567	0.03	0.94	0.991	0.947	0.0026	0.92
relax, learned δ and rates	0.900	0.107	0.048	0.98	0.999	0.991	0.067	0.96
relax, post-LN injection	0.816	0.474	6.5	1.00	0.978	0.845	1.6	1.00
standard block, post-LN injection	0.799	0.511	5	1.00	0.997	0.978	1.8	1.00
relax, $\delta = 1/16$, post-LN injection	0.803	0.230	0.52	0.98	0.939	0.272	0.14	0.99
relax + carry	0.801	0.773	0.96	1.00	0.749	0.581	0.2	1.00
standard block + carry	0.787	0.744	0.57	1.00	–	–	–	–
relax, $\delta = 1/16$ + carry	0.783	0.746	0.076	0.98	0.665	0.398	0.024	0.99
kick, positive rates, zero-init	–	–	–	–	0.889	0.607	0.041	0.87
kick, positive rates + carry	0.907	0.902	0.015	0.63	0.999	0.997	0.05	0.75
relax, $\delta = 1/4$ + carry	0.937	0.939	0.0015	0.83	0.999	0.942	0.0021	0.92

Table 2: All configurations, seed 1238. $\|\Delta s\|_{128}$: step size at $r = 128$; ρ_{128}: shift-free spectral radius at $r = 128$. Groups: unconstrained; constrained damping; post-LN injection $\mathrm{Attn}(\mathrm{LN}(H) + e)$, a washout control; carry, alone and with constraints.

5 Limitations and next steps

- **Scale.** The models are tiny and the tasks are synthetic. Most ablations use one seed; Table 1 has up to three.
- **Contraction versus reversibility.** Contraction and numerically exact inversion pull against each other. Since $|\det D\Phi^r| = e^{-rT\sum(\gamma_i+\alpha_i)}$,

$$\|(D\Phi^r)^{-1}\| \geq e^{\,r\sum_i(\gamma_i+\alpha_i)/(2d)},$$

  so memory-free reversible backpropagation through many iterations of a strongly damped loop is not usable. The detached carry sidesteps this.
- **Length generalisation** (more hops than seen in training) is untested. A finite-state-composition task with no positional embeddings is implemented for it.
- **Untested alternatives.** Over-relaxation (retained fractions in $(-1, 0)$, the positive-momentum sign of relax read as heavy ball), momentum around the whole loop, and halting on the shift-free step are all compatible with exact inversion and remain to be tested.

Related work.

- Looped and recurrent-depth reasoning: Kohli et al. [4]; Liang et al. [5].
- Stabilising the recurrence: STARS [6] regularises the Jacobian spectral radius instead.
- Attractors and the detached carry: Equilibrium Reasoners [2].
- The momentum view of the two-stream couplings: YuriiFormer [7]; Momentum ResNets [8].

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